

Back Paper Examination

Classical Mechanics

Instructor: Prabuddha Chakraborty (prabuddha@isibang.ac.in)

Duration: 120 minutes.

Total points: 50.

Please give arguments where necessary. If it is unclear from your answer why a particular step is being taken, full credit will not be awarded. Grades will be awarded not only based on what final answer you get, but also on the intermediate steps.

1. Consider an equilateral triangle (a two dimensional surface) with uniform mass density. The sides have length l in the beginning. In the process of creating a fractal surface, the equilateral triangle is divided into four smaller equilateral triangles of side length $l/2$ each. Then the middle triangle (one that is formed by joining the midpoints of the sides of the original triangle) is hollowed out. Now, we are left with three equilateral triangles of side length $l/2$ each. This process is carried out again and again infinitely many times, thus forming a fractal surface of total final mass M . Find the moment of inertia of the final object about an axis perpendicular to the original surface and passing through the centroid of the original equilateral triangle.

10 points

2. Imagine a hemispherical bowl of radius R which lies on a surface on its rim. A solid ball of radius r ($r \ll R$) starts from the top of the bowl and rolls down the surface without slipping. At what angle (as measured from the vertical) does the ball roll off the surface of the bowl?

10 points

3. A horizontal stick of uniform density and total mass M and total length L is screwed to a table through its center of mass. It is free to rotate about its center of mass though the center of mass itself remains stationary. Initially the entire stick is at rest. A point mass m , moving horizontally, hits one end of the stick perpendicularly with speed v_0 and sticks to it. Using various conservation laws, describe the motion of the system after the collision. Ignore any force exerted by the table itself on the stick or the point mass.

10 points

4. Consider the double pendulum. A massless stick of length l_1 is pivoted

to a point at the top and a mass m_1 is attached to its lower end. To this mass, is attached another massless stick of length l_2 and to its end, is attached a mass m_2 . The system moves in a single vertical plane under gravity.

- (a) Write down the Lagrangian of the system.
- (b) Derive the equations of motion of the system with the generalized co-ordinates taken as the angle that l_1 makes with the vertical, and the angle that l_2 makes with l_1 .
- (c) Discuss the motion of the system in the two cases:
 - $m_1 \gg m_2$
 - $m_2 \gg m_1$

$$5 + 6 + (2 + 2) = 15$$

5. Show, using a variational method, that the shortest distance between any two points on the surface of a sphere is along the great circle joining the two points. State your assumptions clearly when formulating the calculation.
5 points